CHAPTER IV

Methodology

4.1 Requirements Analysis and Requirements Documentation

Requirements analysis identified three actors: the municipal administrator, household user, and smart meter. The administrator manages system operations, the household monitors consumption and billing, and the smart meter securely transmits water readings to the server.

Use Case Diagram

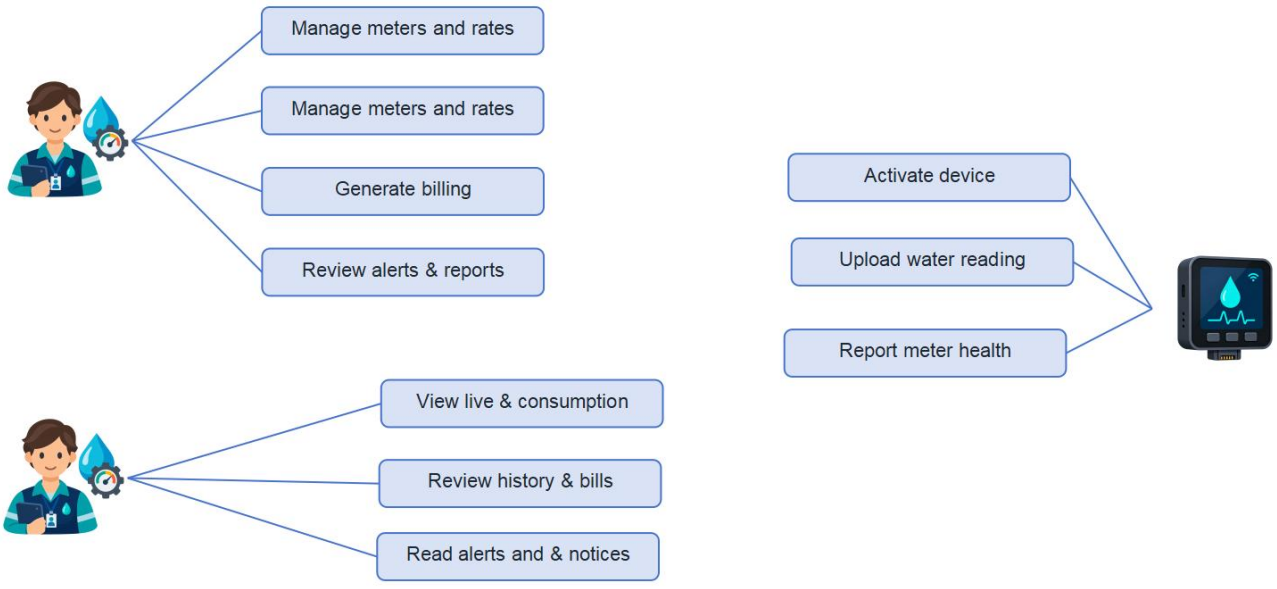


Figure 4.1. Use-case overview of the Smart Water Monitoring Consumption System.

Use Case Desciption

Use Case Name: Manage Households	ID:1	Important Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential, Detail	
Scenario: The administrator wants to create, review, update, or change the operational status of a household account.		
Brief Description: Maintains household identity, service address, contact information, account number, access status, and meter-assignment readiness		
Trigger: The administrator selects Customer Management from the administrative workspace.		
Normal Flow of Events: 1. The administrator signs in and opens Customer Management. 2. The system displays searchable and filterable household records. 3. The administrator creates a new household or selects an existing record. 4. The administrator enters or updates the permitted household information. 5. The system validates required fields, unique account details, role, and status. 6. The system saves the household record and logs the administrative activity.		
Sub Flows:		
Alternative/Exceptional Flows:		

Table 4.1.1 Manage Households

<table>
<tr><td>Use Case Name:
Manage Meters and Rates</td><td>ID:2</td><td>Important Level:
High</td></tr>
<tr><td>Primary Actor:
Municipal Administrator</td><td colspan="2">Use Case Type: Essential, Detail</td></tr>
<tr><td colspan="3">Scenario: The administrator wants to manage municipal smart meters and configure the water rates and monitoring rules applied to households.</td></tr>
<tr><td colspan="3">Brief Description: Combines meter registration, assignment, maintenance, credential control, and effective-dated water-rate and alert configuration.</td></tr>
<tr><td colspan="3">Trigger:The administrator opens Device Management or Water Rate Management.</td></tr>
<tr><td colspan="3">Normal Flow of Events:

1. The administrator opens the required meter or rate module.
2. The system displays the current devices or active billing configurations.
3. For a meter, the administrator registers, assigns, reassigns, suspends, rotates credentials, or decommissions the asset.
4. For a rate, the administrator enters the water rate, fixed charge, subsidy, currency, effective date, and alert thresholds.
5. The system validates identifiers, household eligibility, numeric ranges, and effective dates.
6. The system saves the approved meter or rate configuration and records the activity.</td></tr>
<tr><td colspan="3">Sub Flows:</td></tr>
<tr><td colspan="3">Alternative/Exceptional Flows:</td></tr>
</table>

Table 4.1.2 Manage Meters and Rates

Use Case Name: Generate Billing	ID:3	Important Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential, Detail	
Scenario:The administrator wants to generate a monthly billing statement for an eligible household.		
Brief Description: Aggregates billable consumption, applies the applicable rate configuration, and preserves a historical charge snapshot.		
Trigger: The administrator chooses Generate Statement in Billing Management.		
Normal Flow of Events: 1. The administrator opens Billing Management. 2. The administrator selects the household and billing month. 3. The system validates the period, household, billable readings, and applicable rate. 4. The billing service totals liters and converts the result to cubic meters. 5. The service applies the water rate, fixed charge, subsidy, and opening balance. 6. The system stores the statement, charge breakdown, remaining balance, and payment status. 7. The administrator reviews the generated billing record.		
Sub Flows:		
Alternative/Exceptional Flows:		

Table 4.1.3 Generate Billing

Use Case Name: Review Alerts and Reports	ID:4	Important Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential, Detail	
Scenario:The administrator wants to review leak or consumption alerts and examine municipal operational reports.		
Brief Description: Supports alert investigation, incident follow-up, analytics, filtered reports, and supported data exports.		
Trigger: The administrator opens Leak Monitoring, Reports, Analytics, or the administrative notification view.		
Normal Flow of Events: 1. The administrator selects the alert or report module. 2. The administrator chooses the affected household, report type, date range, or filter. 3. The system validates authorization and input parameters. 4. The system retrieves the related readings, alerts, billing information, and operational summaries. 5. The administrator reviews the evidence and records follow-up notes when an incident is involved. 6. The administrator resolves, marks false, or reopens an alert, or exports a supported report.		
Sub Flows:		
Alternative/Exceptional Flows:		

Table 4.1.4 Review Alerts and Reports

Use Case Name: Review Alerts and Reports	ID:5	Important Level: High
Primary Actor: Household User	Use Case Type: Essential, Detail	
Scenario:The household user wants to view the latest water consumption and meter state.		
Brief Description: Displays the latest owned reading, current flow, interval consumption, estimated billing state, leak state, and meter availability.		
Trigger: The household user opens the Dashboard or Live Monitoring page.		
Normal Flow of Events: 1. The household user signs in. 2. The user opens Dashboard or Live Monitoring. 3. The browser requests the latest household-scoped readings. 4. The server verifies ownership and retrieves the current meter and consumption state. 5. The interface displays current flow, liters used, estimated bill, leak status, meter connection, and recent activity. 6. The interface refreshes according to the configured near-real-time interval.		
Sub Flows:		
Alternative/Exceptional Flows:		

Table 4.1.5 View Live Consumption

Use Case Name: Review History and Bills	ID:6	Important Level: High
Primary Actor: Household User	Use Case Type: Essential, Detail	
Scenario:The household user wants to review past consumption and current or finalized billing information..		
Brief Description: Combines daily, weekly, monthly, or custom consumption history with estimated billing details and monthly statements.		
Trigger: The user opens Consumption History or Estimated Billing		
Normal Flow of Events: 1. The user selects the consumption-history or billing page. 2. For history, the user selects a period or valid custom date range. 3. The server retrieves only the household's readings and computes totals, comparisons, and chart data. 4. For billing, the server selects the applicable rate and eligible consumption information. 5. The interface displays usage summaries, trends, the estimated charge breakdown, and finalized statement history. 6. The user reviews the selected history or billing record.		
Sub Flows:		
Alternative/Exceptional Flows:		

Table 4.1.6 Review History and Bills

Use Case Name: Read Alerts and Notices	ID:7	Important Level: High
Primary Actor: Household User	Use Case Type: Essential, Detail	
Scenario:The household user wants to read leak, consumption, billing, meter, or account notifications.		
Brief Description: Presents household-scoped alerts and official notices while preserving the municipality's source and delivery records.		
Trigger: The user opens Notifications or selects a recent alert from the household workspace.		
Normal Flow of Events: 1. The household user signs in and opens Notifications. 2. The system retrieves notices owned by the household account. 3. The interface displays each notice's type, title, message, date, and read state. 4. The user opens a selected alert or notice. 5. The system marks the notice as read when applicable. 6. The user may remove the notice from the household view.		
Sub Flows:		
Alternative/Exceptional Flows:		

Table 4.1.7 Read Alerts and Notices

Use Case Name: Activate Device	ID:8	Important Level: High
Primary Actor: Smart Meter / IoT Device	Use Case Type: Essential	
Scenario: A registered M5Stack smart meter wants to exchange its one-time code for a scoped device credential.		
Brief Description: Binds the physical hardware to the municipal device record and activates authenticated telemetry access.		
Trigger: The technician enters the administrator-issued activation code on the registered device.		
Normal Flow of Events: 1. The administrator has already registered the meter and issued a short-lived activation code. 2. The device submits its UUID, activation code, hardware identity, and firmware version. 3. The server verifies that the code is valid, unused, and unexpired. 4. The server checks device status and hardware-binding rules. 5. The server binds the hardware identity and marks the device active. 6. The server issues a scoped device credential. 7. The M5Stack stores the credential and becomes ready to upload telemetry.		
Sub Flows:		
Alternative/Exceptional Flows:		

Table 4.1.8 Activate Device

Use Case Name: Upload Water Reading	ID:9	Important Level: High
Primary Actor: Smart Meter / IoT Device	Use Case Type: Essential	
Scenario: An activated smart meter wants to submit a measured water-consumption interval..		
Brief Description: Authenticates the device, validates telemetry, stores the reading once, and returns authoritative totals and alert state.		
Trigger: The firmware reaches the configured upload interval with a pending measurement.		
Normal Flow of Events: 1. The sensor counts valid flow pulses during the sampling interval. 2. The firmware calculates flow rate and interval liters. 3. The device creates a stable reading identifier and stores the pending state. 4. The meter submits the reading with its scoped credential. 5. The server derives the device and household from the authenticated identity. 6. The server validates and stores the reading idempotently. 7. The server evaluates thresholds and returns the authoritative total and leak state. 8. After acknowledgement, the meter clears the pending interval and advances its sequence.		
Sub Flows:		
Alternative/Exceptional Flows:		

Table 4.1.9 Upload Water Reading

Use Case Name: Report Meter Health	ID:10	Important Level: Medium
Primary Actor: Smart Meter / IoT Device	Use Case Type: Supporting	
Scenario: The smart meter wants to report its connection, firmware, and operational health to the server.		
Brief Description: Updates municipal visibility of device connectivity, last-seen activity, telemetry status, firmware version, and relevant diagnostics.		
Trigger: The device activates, uploads telemetry, reconnects to the network, or reaches a health-reporting interval.		
Normal Flow of Events: 1. The M5Stack evaluates its Wi-Fi and server connection state. 2. The device prepares its hardware identity, firmware version, monitoring state, signal strength, and health metadata. 3. The device submits the health information using its scoped credential. 4. The server validates the authenticated device identity. 5. The server updates last-seen, last-telemetry, firmware, and operational health fields. 6. The administrator workspace displays the latest device status.		
Sub Flows:		
Alternative/Exceptional Flows:		

Table 4.1.10 Report Meter Health

4.2 System Design, Software, and Processes

The system uses an Internet of Things client-server design. The physical meter performs measurement and short-term persistence, while the Laravel server remains authoritative for identity, ownership, cumulative usage, billing rules, threshold evaluation, and record storage. The browser interfaces consume protected JSON endpoints and present role-specific workspaces.

Output and User Interface Design

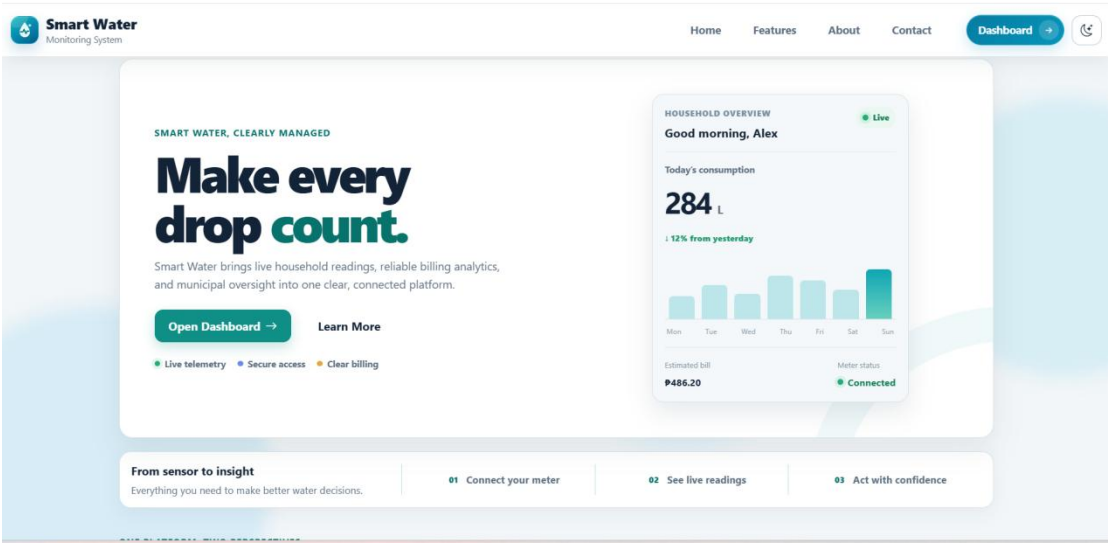


Figure 4.2.1 Home Interface

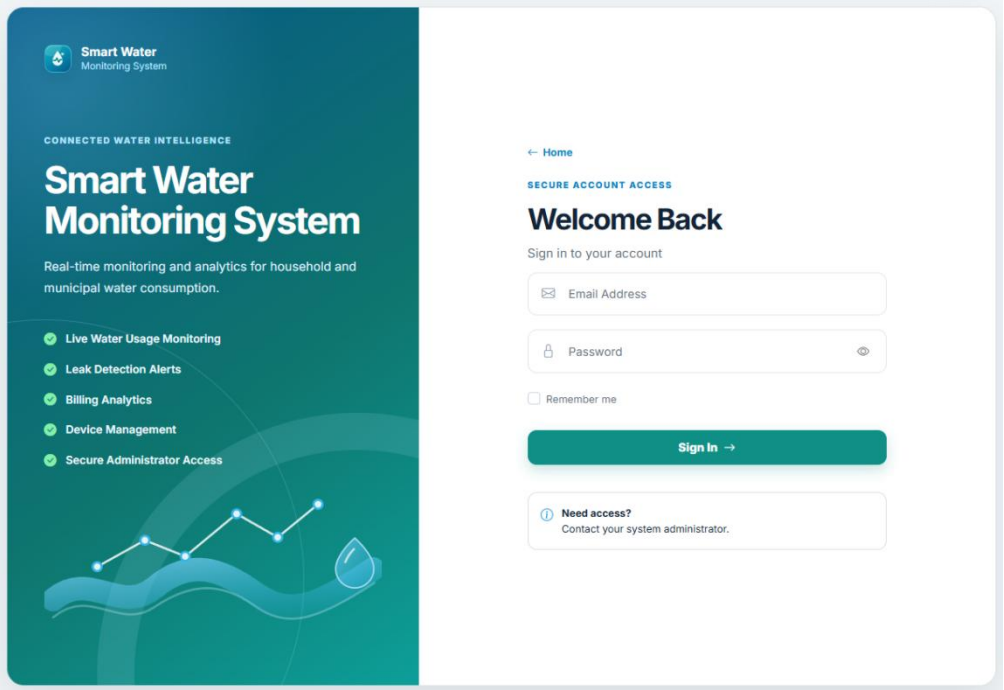


Figure 4.2.2 Log in Interface

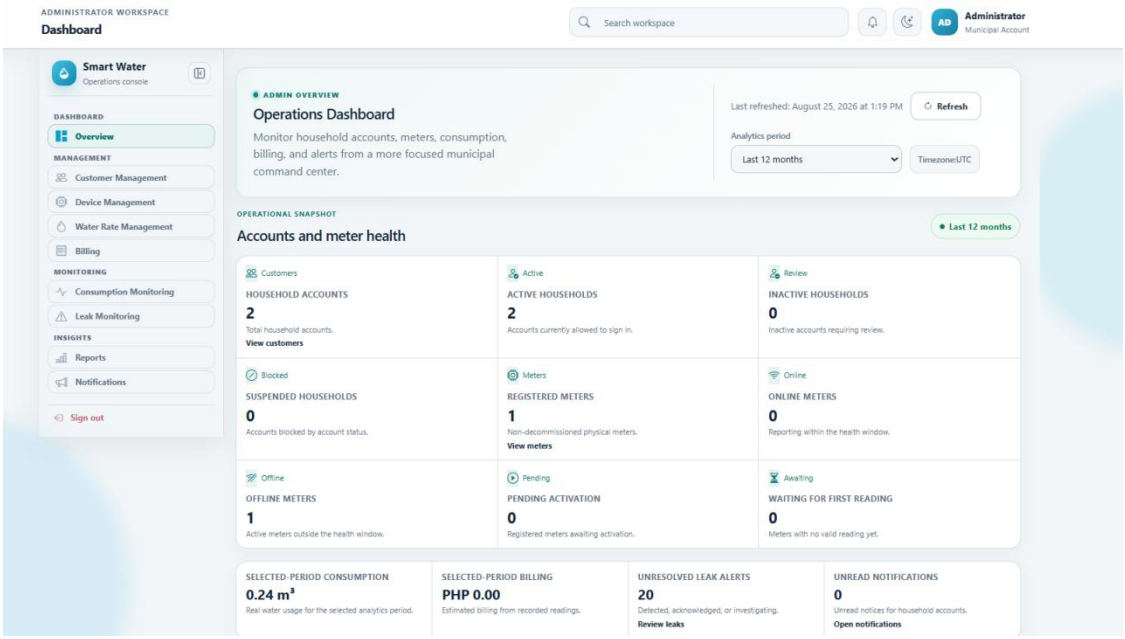


Figure 4.2.3 Admin (Dashboard) Interface

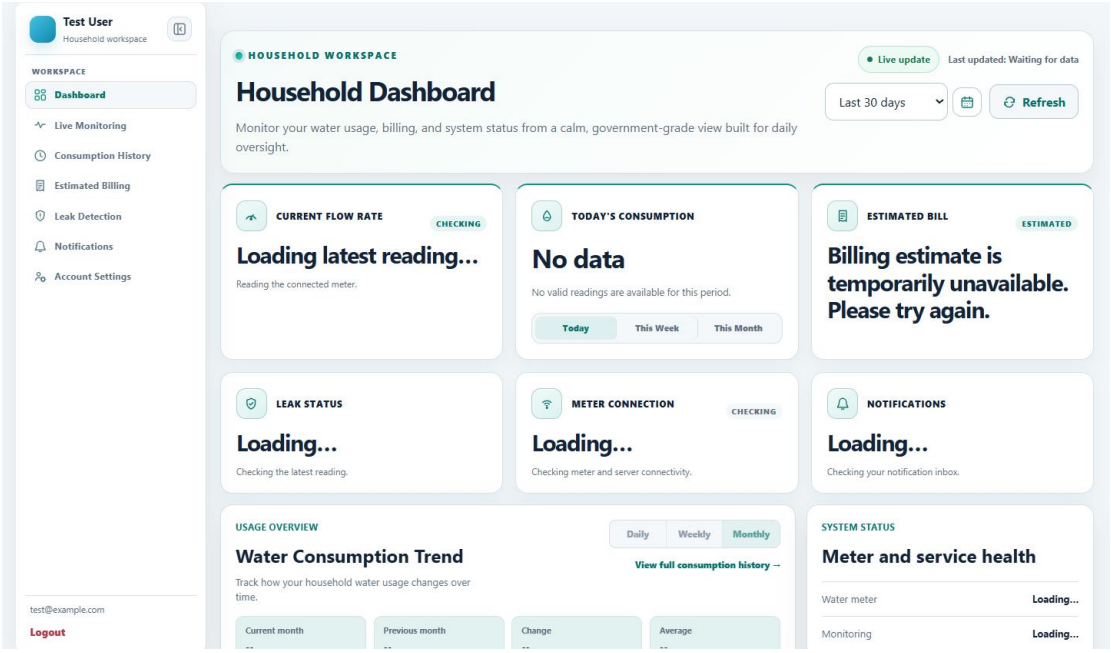
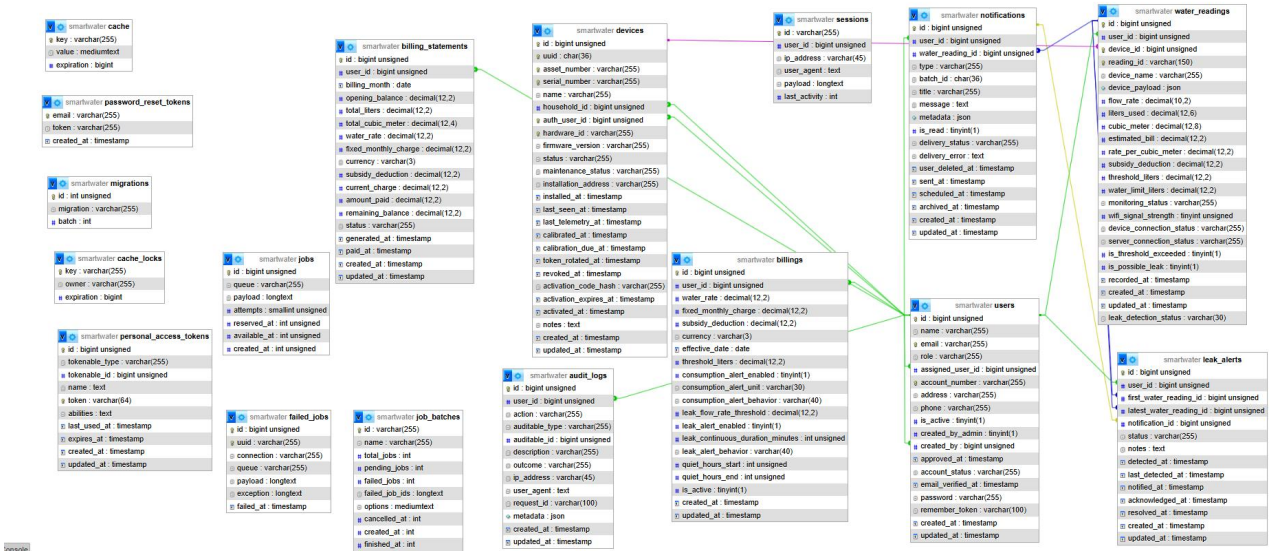


Figure 4.2.4 Household (Dashboard) Interface

4.3 Entity Relationship Diagram



4.4 Data Dictionary

The following data dictionary summarizes the operational tables most directly involved in monitoring, billing, alerting, and administration. Framework-support tables for cache, jobs, sessions, password resets, and personal access tokens remain part of the implementation but are omitted from the main discussion for clarity.

Field Name	Data Type	Length	Description
id	BIGINT	20	
name	VARCHAR	255	
email	VARCHAR	255	
role	VARCHAR	255	
account_number	VARCHAR	255	
address	VARCHAR	255	

phone	VARCHAR	255	
is_active	BOOLEAN	1	
email_verified_at	DATETIME	-	
password	VARCHAR	255	
remember_token	VARCHAR	100	
created_by_admin	BOOLEAN	1	
assigned_user_id	BIGINT	20	
created_by	BIGINT	20	
approved_at	DATETIME	-	
account_status	VARCHAR	255	
created_at	DATETIME	-	
updated_at	DATETIME	-	

Table 4.4.1 Database Table of Users

Field Name	Data Type	Length	Description
id	BIGINT	20	
uuid	VARCHAR	255	
asset_number	VARCHAR	255	
serial_number	VARCHAR	255	
name	VARCHAR	255	
household_id	BIGINT	20	
auth_user_id	BIGINT	20	
hardware_id	VARCHAR	255	
firmware_version	VARCHAR	255	
status	VARCHAR	255	
maintenance_status	VARCHAR	255	
installation_address	VARCHAR	255	
installed_at	DATETIME	-	
last_seen_at	DATETIME	-	
last_telemetry_at	DATETIME	-	
calibrated_at	DATETIME	-	
calibration_due_at	DATETIME	-	
token_rotated_at	DATETIME	-	
revoked_at	DATETIME	-	
activation_code_hash	VARCHAR	255	

activation_expires_at	DATETIME	-	
activated_at	DATETIME	-	
notes	TEXT	-	
created_at	DATETIME	-	
updated_at	DATETIME	-	

Table 4.4.2 Database Table of Devices

Field Name	Data Type	Length	Description
id	BIGINT	20	
user_id	BIGINT	20	
device_id	BIGINT	20	
reading_id	VARCHAR	255	
flow_rate	DECIMAL	18,6	
liters_used	DECIMAL	18,6	
cubic_meter	DECIMAL	18,6	
estimated_bill	DECIMAL	12,2	
rate_per_cubic_meter	DECIMAL	12,4	
subsidy_deduction	DECIMAL	12,2	
threshold_liters	DECIMAL	18,6	
is_threshold_exceeded	BOOLEAN	1	
is_possible_leak	BOOLEAN	1	
leak_detection_status	VARCHAR	255	
water_limit_liters	DECIMAL	18,6	
monitoring_status	VARCHAR	255	
wifi_signal_strength	INTEGER	11	
device_connection_status	VARCHAR	255	
server_connection_status	VARCHAR	255	
device_name	VARCHAR	255	
device_payload	JSON/TEXT	-	
recorded_at	DATETIME	-	
created_at	DATETIME	-	
updated_at	DATETIME	-	

Table 4.4.3 Database Table of Water Readings

Field Name	Data Type	Length	
id	BIGINT	20	
user_id	BIGINT	20	
water_rate	DECIMAL	12,4	
fixed_monthly_charge	DECIMAL	12,2	
subsidy_deduction	DECIMAL	12,2	
currency	VARCHAR	3	
effective_date	DATE	-	
threshold_liters	DECIMAL	18,6	
consumption_alert_enabled	BOOLEAN	1	
consumption_alert_unit	VARCHAR	255	
consumption_alert_behavior	VARCHAR	255	
leak_alert_enabled	BOOLEAN	1	
leak_flow_rate_threshold	DECIMAL	18,6	
leak_continuous_duration_minutes	INTEGER	11	
leak_alert_behavior	VARCHAR	255	
quiet_hours_start	INTEGER	2	
quiet_hours_end	INTEGER	2	
is_active	BOOLEAN	1	
created_at	DATETIME	-	
updated_at	DATETIME	-	

Table 4.4.4 Database Table of Billing Configurations

Field Name	Data Type	Length	Description
id	BIGINT	20	
user_id	BIGINT	20	
billing_month	DATE	-	
opening_balance	DECIMAL	12,2	
total_liters	DECIMAL	18,6	
total_cubic_meter	DECIMAL	18,6	
water_rate	DECIMAL	12,4	

fixed_monthly_charge	DECIMAL	12,2	
subsidy_deduction	DECIMAL	12,2	
currency	VARCHAR	3	
current_charge	DECIMAL	12,2	
amount_paid	DECIMAL	12,2	
remaining_balance	DECIMAL	12,2	
status	VARCHAR	255	
generated_at	DATETIME	-	
paid_at	DATETIME	-	
created_at	DATETIME	-	
updated_at	DATETIME	-	

Table 4.4.5 Database Table of Billing Statements

Field Name	Data Type	Length	Description
id	BIGINT	20	
user_id	BIGINT	20	
water_reading_id	BIGINT	20	
type	VARCHAR	255	
title	VARCHAR	255	
message	TEXT	-	
metadata	JSON/TEXT	-	
is_read	BOOLEAN	1	
batch_id	VARCHAR	255	
delivery_status	VARCHAR	255	
delivery_error	TEXT	-	
scheduled_at	DATETIME	-	
sent_at	DATETIME	-	
archived_at	DATETIME	-	
user_deleted_at	DATETIME	-	
created_at	DATETIME	-	
updated_at	DATETIME	-	

Table 4.4.6 Database Table of Notifications

Field Name	Data Type	Length	Description
id	BIGINT	20	
user_id	BIGINT	20	
first_water_reading_id	BIGINT	20	
latest_water_reading_id	BIGINT	20	
notification_id	BIGINT	20	
status	VARCHAR	255	
notes	TEXT	-	
detected_at	DATETIME	-	
last_detected_at	DATETIME	-	
notified_at	DATETIME	-	
acknowledged_at	DATETIME	-	
resolved_at	DATETIME	-	
created_at	DATETIME	-	
updated_at	DATETIME	-	

Table 4.4.7 Database Table of Leak Alerts

Field Name	Data Type	Length	Description
id	BIGINT	20	
user_id	BIGINT	20	
action	VARCHAR	255	
auditable_type	VARCHAR	255	
auditable_id	BIGINT	20	
description	VARCHAR	255	
metadata	JSON/TEXT	-	
outcome	VARCHAR	255	
ip_address	VARCHAR	45	
user_agent	TEXT	-	
request_id	VARCHAR	255	
created_at	DATETIME	-	
updated_at	DATETIME	-	

Table 4.4.8 Database Table of Audit Logs

4.5 Development and Testing

Continuous development and testing were conducted to ensure the reliability of the Smart Water Monitoring System. The Laravel application, database, M5Stack CoreS3, and YF-S201 water-flow sensor were tested to verify pulse detection, data transmission, billing calculations, alerts, and user management. Identified errors were corrected before implementation.

4.6 Implementation Plan

The implementation plan provides a structured process for deploying and operating the system. It includes configuring the water-flow sensor and CoreS3, connecting the device to Wi-Fi, preparing the Laravel server and database, registering users and devices, and verifying that readings appear correctly on the household and administrator dashboards.

Implementation Issues and Challenges

Challenges encountered included incorrect wiring, defective sensor components, unstable signal readings, network connection errors, and inaccurate flow-rate displays. These issues were addressed through hardware testing, resistor replacement, wiring correction, software debugging, database validation, and repeated water-flow tests.

4.7 System Implementation and Results

The Smart Water Monitoring System was successfully implemented using the M5Stack CoreS3, YF-S201 sensor, Laravel application, and MySQL database. The sensor detects water-flow pulses, while the CoreS3 calculates

consumption and transmits readings to the server. The system records usage, displays current and historical data, estimates billing, generates alerts, and provides administrative monitoring. Testing confirmed that the application, database relationships, and device-to-server communication function as intended.